Economics S5040: Dynamic Stochastic Modeling
Lecture 19 — Stochastic Discount Factors in a Markov Economy

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Central question: Why can two assets with the same expected payoff have different prices? Because payoffs are more valuable when they arrive in states in which consumption is scarce and marginal utility is high.

Plan for an 80-minute class

Minutes	Topic
0–10	From intertemporal choice to asset pricing
10–28	Deriving the stochastic discount factor
28–43	Finite-state Markov representation
43–58	State prices and the pricing matrix
58–70	State-dependent risk-free rates
70–80	Risk, covariance, and economic interpretation

1 The household's asset-pricing problem

Consider a representative household with preferences

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \qquad 0 < \beta < 1.$$

Suppose an asset costs p_t units of consumption today and pays x_{t+1} units of consumption next period. Let θ_t denote the number of units purchased.

A marginal increase in θ_t lowers current consumption by p_t and raises next-period consumption by x_{t+1}. The first-order condition is

$$u'(c_t)p_t = \beta \mathbb{E}_t \left[u'(c_{t+1})x_{t+1} \right].$$

Dividing by current marginal utility gives

$$\boxed{p_t = \mathbb{E}_t[m_{t+1}x_{t+1}],}$$

where

$$\boxed{m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}}$$

is the **stochastic discount factor** (SDF).

Key economic idea. The SDF converts a payoff in a future state into its value in units of consumption today. It is large when next-period marginal utility is high.

2 Why the discount factor is stochastic

If consumption tomorrow were known with certainty, then m_{t+1} would be known today. Under consumption risk, however, marginal utility tomorrow depends on the realized state.

With CRRA utility,

$$u(c) = \begin{cases} \dfrac{c^{1-\rho} - 1}{1-\rho}, & \rho \neq 1, \\ \log c, & \rho = 1, \end{cases} \qquad u'(c) = c^{-\rho}.$$

Hence

$$\boxed{m_{t+1} = \beta \left(\frac{c_{t+1}}{c_t} \right)^{-\rho}.}$$

If consumption growth is low, m_{t+1} is high. A payoff in that state is valuable because it arrives precisely when consumption is scarce.

For log utility ($\rho = 1$),

$$m_{t+1} = \beta \frac{c_t}{c_{t+1}}.$$

3 A finite-state Markov economy

Let the state be

$$s_t \in \{1, 2, \ldots, N\},$$

with transition matrix

$$P = [P_{ij}], \qquad P_{ij} = \Pr(s_{t+1} = j \mid s_t = i).$$

Consumption depends only on the current state:

$$c_t = c_i \quad \text{when } s_t = i.$$

If the current state is i and next period's state is j, the SDF is

$$\boxed{m_{ij} = \beta \frac{u'(c_j)}{u'(c_i)}.}$$

For CRRA utility,

$$\boxed{m_{ij} = \beta \left(\frac{c_j}{c_i} \right)^{-\rho}.}$$

Thus the Markov chain tells us *how likely* each state is, while the SDF tells us *how valuable* consumption is in each state.

4 A three-state classroom example

Consider

$$c = \begin{pmatrix} 0.95 \\ 1.00 \\ 1.05 \end{pmatrix}, \qquad P = \begin{pmatrix} 0.80 & 0.15 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.15 & 0.80 \end{pmatrix}.$$

Interpret the states as low, normal, and high aggregate consumption. Let

$$\beta = 0.96, \qquad \rho = 2.$$

Then

$$m_{ij} = 0.96 \left(\frac{c_j}{c_i} \right)^{-2}.$$

Ask the class before calculating:

> Starting from the normal state, which one-dollar payoff is worth more: a dollar delivered in the low-consumption state or a dollar delivered in the high-consumption state?

The answer is the low-consumption payoff. Its marginal utility weight is larger.

5 State prices

Suppose a security pays one unit of consumption next period only if state j occurs. Its price in current state i is

$$q_{ij} = P_{ij} m_{ij}.$$

Define the **pricing matrix**

$$\boxed{Q = [q_{ij}] = P \circ M,}$$

where $\circ$ denotes element-by-element multiplication and $M = [m_{ij}]$.

The distinction is important:

- P_{ij} is a physical probability.
- m_{ij} is a marginal-utility discount factor.
- $q_{ij} = P_{ij} m_{ij}$ is the current price of one unit of consumption delivered only in state j next period.

If a one-period asset has state-contingent payoff vector

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix},$$

then its price vector is simply

$$\boxed{p = Qx.}$$

The ith entry is

$$p_i = \sum_j q_{ij} x_j = \sum_j P_{ij} m_{ij} x_j.$$

6 The risk-free asset

A one-period risk-free bond pays one unit of consumption in every next-period state. Its payoff vector is

$$\mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Therefore its price in each current state is

$$\boxed{q^f = Q\mathbf{1}.}$$

Equivalently,

$$q_i^f = \sum_j P_{ij} m_{ij} = \mathbb{E}_i[m_{t+1}].$$

The gross risk-free return is

$$\boxed{R_i^f = \frac{1}{q_i^f}.}$$

Notice that R_i^f can vary with the current state even though the payoff is risk free. What is risk free is the *next-period payoff*, not necessarily the price at which the bond is purchased today.

> **Important distinction.** A risk-free bond has the same payoff in every next-period state, but its price can depend on today's state because today's marginal utility and the distribution of tomorrow's consumption depend on today's state.

7 Returns and the fundamental pricing equation

If an asset has price p_t and payoff x_{t+1}, define its gross return by

$$R_{t+1} = \frac{x_{t+1}}{p_t}.$$

Then

$$p_t = \mathbb{E}_t[m_{t+1} x_{t+1}]$$

implies

$$\boxed{1 = \mathbb{E}_t[m_{t+1} R_{t+1}].}$$

For the risk-free asset,

$$1 = R_t^f \mathbb{E}_t[m_{t+1}],$$

so again

$$R_t^f = \frac{1}{\mathbb{E}_t[m_{t+1}]}.$$

8 Expected returns and covariance

Using

$$\mathbb{E}[mR] = \mathbb{E}[m]\mathbb{E}[R] + \mathrm{Cov}(m, R),$$

we obtain

$$1 = \mathbb{E}_t[m_{t+1}]\mathbb{E}_t[R_{t+1}] + \mathrm{Cov}_t(m_{t+1}, R_{t+1}).$$

Since $R_t^f = 1/\mathbb{E}_t[m_{t+1}]$,

$$\boxed{\mathbb{E}_t[R_{t+1}] - R_t^f = -R_t^f \,\mathrm{Cov}_t(m_{t+1}, R_{t+1}).}$$

This equation gives the economic meaning of a risk premium.

- If an asset pays well when m is high—bad consumption states—it provides insurance. Its covariance with m is positive, so investors are willing to accept a lower expected return.
- If an asset pays poorly when m is high, it performs badly exactly when consumption is scarce. Its covariance with m is negative, so it must offer a higher expected return.

9 What is "risk" in consumption-based asset pricing?

Risk is not simply the variance of an asset's payoff. What matters is how the payoff covaries with marginal utility.

Two assets can have the same expected payoff and even similar payoff variance, but different prices if one pays in bad aggregate states and the other pays in good aggregate states.

Asset-pricing principle:

A payoff is valuable when it arrives in a state in which marginal utility is high.

10 Connection to earlier weeks

The structure should now look familiar:

$$\text{Markov states} \longrightarrow P \longrightarrow \text{expectations over future states.}$$

What is new is the marginal-utility weight:

$$P_{ij} \longrightarrow P_{ij} m_{ij} = q_{ij}.$$

The transition matrix tells us where the economy may go. The pricing matrix tells us how much a unit of consumption in each possible destination is worth today.

Summary

1. Optimal intertemporal choice implies
$$p_t = \mathbb{E}_t[m_{t+1} x_{t+1}].$$

2. The stochastic discount factor is
$$m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}.$$

3. In a finite-state Markov economy,
$$q_{ij} = P_{ij} m_{ij}, \qquad Q = P \circ M.$$

4. One-period asset prices satisfy
$$p = Qx.$$

5. The risk-free bond price and return are
$$q^f = Q\mathbf{1}, \qquad R_i^f = 1/q_i^f.$$

6. Expected return premia depend on covariance with marginal utility, not simply payoff variance.

Preview of Lecture 20. Today we priced one-period claims. Next time we price an asset that survives into the future. Its price will satisfy a recursive equation:
$$v = Q(d + v).$$
This is the asset-pricing analogue of the recursive equations used throughout the course.